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ORIGINAL ARTICLE

Facial rejuvenation with botulinum neurotoxin: An anatomical and experiential perspective

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Abstract
The utilization of botulinum neurotoxin for facial rejuvenation has been progressively expanding into multiple upper, mid, and lower facial and neck applications. Success in these areas is dependent upon a number of factors, above all a sophisticated understanding of the underlying facial anatomy. In addition, although published consensus recommendations can guide dosing and injection site placement, expert technical application is critical, as is individualized treatment for each patient, based on an aesthetic assessment. This includes an evaluation of the patient's unique functional anatomy and a frank discussion of the patient's desires and goals. Only by addressing these considerations can the best possible outcomes be consistently achieved.

Key words: *botulinum neurotoxin type A, facial rejuvenation, functional anatomy, injection technique, aesthetic assessment*

If one looks at any attractive visage, similarities exist in terms of smooth, round contours, high cheekbones, oblique, hollow jowls, and a thin but well defined jawline. We refer to such features as the ''triangle of beauty''. It is thought that the triangle of beauty creates desire on an unconscious level by encompassing two vertical oblique vectors that open to the sky.

The opposite of the triangle of beauty, the reverse triangle, can be found in faces considered to be old or unattractive. In this reverse triangle, the summit is on top and the base is below, with specific components that include drooping eyes, lateral eyebrow ptosis, malar descent, a large, jowly, heavy jawline, and hypertonic contractions of the depressor muscles (which pull down the soft facial components).

Approaching the topic of facial rejuvenation with a patient should begin with a discussion of beauty and aging and the specific goals desired. Usually, it is a look that is ''natural'', not ''surgical''. To obtain a natural, physiological correction, we must correct properly—step-by-step—each specific anatomical component. Harmony is the key to success in creating a natural look versus one that is excessively modified and ultimately unnatural.

In our opinion, the best way to approach facial rejuvenation in an elegant manner is to learn the anatomy of the face again in detail in order to achieve a comprehensive knowledge. Both dynamic and volumetric anatomical changes need to be thoroughly understood and addressed (1).

Dynamic changes are those resulting from the utilization of the facial musculature. Such changes can be modified with products that relax lines and smooth and shape the face—botulinum neurotoxin type A being a popular choice. Botulinum neurotoxin type A has traditionally been used to improve the aesthetics of the upper face, where dynamic changes are most often apparent. Specifically, botulinum neurotoxin type A has been used to treat lines present in the glabella, forehead, and crow's-feet areas (2,3). Its use, however, is expanding into a number of additional upper, mid, and lower facial and neck applications, alone or in combination with other treatment modalities (e.g. fillers, laser resurfacing). Unless otherwise specified, all discussions of botulinum neurotoxin type A in this article refer to BOTOX® (Allergan, Inc., Irvine, CA, USA), the most widely researched and utilized product available. Botulinum neurotoxin type A agents are not interchangeable.

Volumetric change is another important anatomical issue when discussing facial rejuvenation. Most often seen in the mid and lower face and neck, volumetric changes can be addressed surgically or

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nonsurgically and can be related to one of two distinct diagnoses: 1) a desire for reshaping (in cases with adequate volume), where remodeling of the face is performed in order to produce a return to a more youthful, delicate visage; or 2) a desire for refilling (in cases of soft tissue loss or inadequate volume or to fill wrinkles), for which injectable soft tissue augmentation agents such as dermal fillers or fat grafting can be used.

While it is useful from an academic perspective to look at dynamic versus volumetric changes, the trend currently is toward assessing the face as a whole and using combination therapy (e.g. botulinum neurotoxin type A plus fillers, surgery plus fillers) to provide each patient with the best possible outcomes (1,4).

General considerations for botulinum neurotoxin type A utilization

Botulinum neurotoxin type A injections are the most widely utilized nonsurgical cosmetic procedures in the United States (5). Although botulinum neurotoxin type A traditionally has been employed in the upper face to address dynamic changes, its use is spreading to areas where volumetric changes are more central (e.g. the mid and lower face and neck), often in combination with other modalities.

Key to the achievement of consistent, effective, aesthetically desirable effects with botulinum neurotoxin type A is an in-depth knowledge of functional anatomy as it relates to volume and the three-dimensional nature of the face. Facial muscles are not all flat, nor are they static; all facial anatomical components are in permanent motion. Because muscles provide all movement and change to the face, they can be thought of as the ''heart of the face''.

As muscles lie in different planes, an understanding of muscular topography and depth is necessary. Immediately under the skin is the first group of muscles, known as the superficial muscular aponeurotic system (SMAS), which includes muscles from the same embryologic layer origin. SMAS muscles are very thin (1–2 mm), long and flat, and adhere to the skin. Deep below this superficial layer lie shorter and stronger muscles.

Understanding the location of the bone and skin insertions of each muscle and their variabilities is also imperative. Muscle mass, strength, and activity differ from patient to patient, with expressive people having a more active facial musculature. In some people, the extreme form of this—hyperkineticity, or constant movement of the face—can result in blepharospasm or hemifacial spasm. The degree of tonicity also varies and needs to be analyzed; hypotonic faces, however, differ from those that are hypertonic. Also, many patients display asymmetries that are detectable by analyzing facial expression.

Finally, it is important to remember that there is more to the face than just muscles. Muscles are surrounded by fat and subcutaneous soft tissue and cover a bumpy facial skeleton, which is very different in each patient. Skin, fat, bones, nerves, vessels, and glands all need to be considered (1,6,7). If botulinum neurotoxin type A utilization is to be optimal, clinicians also should conduct a comprehensive patient evaluation. This includes a complete medical and cosmetic history, questions intended to elicit a patient's desires and goals, and an aesthetic evaluation. The latter is particularly important, because the way in which each person uses his or her facial anatomy can differ greatly. Each assessment should include an evaluation of facial expression at rest and during animation, when the range of motion of involved muscles should be evaluated. In addition, each muscle should be palpated (during repose and contraction); and muscle strength, asymmetry, and overall muscular balance should be assessed.

Patient communication is another important component to the successful utilization of botulinum neurotoxin type A. Patients feel more confident if the physician explains a proposed procedure and its justification. Similarly, reasonable expectations regarding outcomes should be set, and potential adverse effects should be explained. One of the many benefits of botulinum neurotoxin type A is that effects are completely reversible with time.

Consensus recommendations regarding dosage, injection sites, and techniques are available and serve as a valuable starting point (2). Expert technical application is also of the utmost importance. However, the importance of individualizing treatment for each patient—based on his or her anatomical assessment and stated goals—cannot be overemphasized. Dosing must be modified and injection points distributed according to observed animation, muscle function, and desired effect.

A final key to successful utilization of botulinum neurotoxin type A is documentation. Recording the specifics of injection for each patient (where, how much) for each procedure in the clinic chart allows for duplicating the procedure when patients have been satisfied with the results; conversely, documentation allows the physician to modify treatment for those in whom outcomes could be improved.

Botulinum neurotoxin type A in the upper face

Botulinum neurotoxin type A efficacy is well established in the upper face in the glabella, forehead, and crow's-feet areas, where dynamic wrinkling is the major sign of aging. Anatomically, these dynamic changes are due to the fact that the upper face is more muscular than the lower face; that muscles, skin, and bone are strongly attached—are very ''closed''—in the upper face; and that there is minimal fat in this area. Thus, immediately after a

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contraction (e.g. a facial expression), the skin shrinks, creating wrinkles.

The superficial muscles in the upper face include the frontalis, procerus, medial orbicularis oculi, lateral orbicularis oculi, and depressor supercilii (the internal part of the orbicularis oculi, under the projection of the head at the eyebrow); the nasalis may also be involved. Deeper muscles that need to be considered include the corrugator supercilii (under the procerus and frontalis; a very deep muscle with both a bone and a skin insertion: the possibility of diffusion into this muscle is important to take into account) and the depressor supercilii (internal to the orbicularis oculi, under the projection of the head at the eyebrow). The zygomaticus major and minor (under the orbicularis oculi) and levator labii superioris and levator labii superioris alaeque nasi (under the orbicularis oris, malar fat pad, and nasolabial folds) may also be involved. Figure 1 shows the planar views of some of the important upper facial muscles relevant to cosmetic botulinum neurotoxin injection (7).

General anatomical and treatment guidelines for the forehead, glabella, and crow's-feet areas are provided in Table I (2,7–10). These are the most common sites of treatment, as botulinum neurotoxin type A use is generally predictable and safe in these areas. Nonetheless, the sites of injection and doses used will vary from patient to patient, depending on factors such as gender, size, skin thickness, muscle mass, and muscle activity in these areas.

In addition to these traditional applications, more advanced botulinum neurotoxin type A applications in the upper face include treatment of dermatochalasis (11), eyebrow shaping (targeting the lateral orbital sub-brow orbicularis to raise the lateral brow and/or central frontalis muscle above the medial eyebrow to depress the medial brow), upper lid ptosis (pretarsal orbicularis oculi), fold asymmetries (treating a variety of the eyebrow depressors including the appropriate portion of orbicularis muscle, frontalis muscle, procerus, and corrugator muscles), infraorbital creases (orbicularis oculi), upper and lower lid shaping (targeted orbicularis oculi muscle), and lower eyelid asymmetries (preseptal orbicularis oculi). Figure 2 provides some suggested injection sites for these advanced botulinum neurotoxin type A applications around the eyes.

When injecting below the eye, a snap test must be performed prior to treatment to avoid the potential for corneal show following treatment (8–10,12). Often in these expanded applications, the agonist muscle is treated, allowing the antagonist muscle to exhibit overactivity. Low doses (~1 U/injection site) are generally recommended for these types of applications. In addition, the sites of injection and doses injected will vary from patient to patient, depending on the specific outcomes desired. The precise placement of injections in these areas becomes an art form in the hands of the experienced injector, targeting the chemodenervation to specific areas in these small muscles. The ability for botulinum neurotoxin type A (i.e. BOTOX) to be safe and efficacious in these areas is a function of its low propensity to diffuse, or migrate, from the site of injection (13).

When treating the upper face with botulinum neurotoxin type A, it is important to be aware of a number of anatomical features in addition to the musculature (6,7,14). In the forehead, for instance, are the frontal branches of the facial nerve in the temporal area; the subcutaneous fat also is very thick in the eyebrow area. When injecting into the glabella, it should be remembered that the



Figure 1. Muscles of the upper face (*corrugator*, *procerus*, *orbicularis oculi*, and *frontalis*) shown in planar sketch and dissection view. In the sketch, the *corrugator* muscle is shown underneath the *orbicularis* and *frontalis* muscles. In the dissection view, the *corrugator* and *procerus* muscles are shown after raising the *orbicularis* and *frontalis* muscles. Note (in blue), crossing the *depressor supercilii* and the *corrugator* is the *supratrochlear* (ST) vein (and ST nerve (in white) and ST artery (in red)). Sketch by G. Blanchet; photograph, courtesy of L. Belhaouari: Belhaouari L, Gassia V. L'Art de la Toxine Botulique [in French]. Editions Arnette (Wolters Kluwer France): Rueil-Malmaison, France, 2006. Reproduced by kind permission of the Publisher (7).

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Table I. Traditional botulinum neurotoxin type A applications: upper face (2,7–10).

Treatment area	Muscles involved	Anatomical considerations	Total BoNTA dose/ injection sites	Injection technique	Notes
Forehead	Frontalis	Frontalis originates on the galea aponeurotica layer superiorly, interdigitates with the procerus, corrugator, and orbicularis oculi muscles overlying the brow area; paired muscles with distinct midline separation; contraction elevates the brow, retracts the scalp; lies beneath a thick layer of skin/subcutaneous tissue	10–20 U (women), 20–30 U (men) 2–8+ sites	Deep injections into the muscle recommended; staggered distribution, ~1 cm distance between injections, positioned 1.5–2 cm above brow margin; injection sites should be lateral enough to avoid a quizzical eyebrow appearance; avoid the lower lateral forehead	Dosing/injection pattern highly variable. In women, the frontalis should be treated in conjunction with the glabella
Glabella	Procerus, corrugator supercilii, depressor supercilii, orbicularis oculi, nasalis (may be involved), levator alaeque nasi (may be involved)	Procerus is a thin, pyramidal muscle, lies vertically in the midline; originates at lower ends of nasal bones; inserts superiorly to soft tissue/frontalis between the eyebrows; contraction causes inferior displacement of medial angle of eyebrow. Corrugator is a paired muscle; medial origin is at the junction of the frontal and nasal bones; inserts on skin in the midbrow region; contraction narrows the eyebrows and depresses brow	5–30+ U (women), 5–40+ U (men) 1–10 sites	Superficial injection of procerus midline below the eyebrows; serial deep puncture technique directly into corrugator; avoid injecting too low over the orbit; in general injections should be directed outside the orbital rim 1 cm above. The injection is deeper, near to the periosteum (bone contact)	Muscular balance must be kept top-of-mind; most common site of BoNTA injection
Crow's-feet	Orbicularis oculi, other muscles may be involved	Circumferential muscle that surrounds the orbital rim, divided into 3 parts: pretarsal, preseptal palpebral, and periorbital; contraction closes the eye; lateral aspect of orbital portion of the muscle should be targeted because the forced maximum contraction pulls down the tail of the eyebrow; veins/vessels plentiful in this area	<12–30+ U 2–5+ sites	Approach the orbital rim so there is no chance of diffusion (the wide orbicularis acts as a barrier against diffusion); injections should be tangential and on a subdermal plane	Patients who recruit the zygomaticus for crow's-feet animation should not be treated with BoNTA

BoNTA=botulinum neurotoxin type A.

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supratrochlear and supraorbital nerves (which are sensitive) and vessels are found in this area (around the gap of the trochlea, pulling on the superior oblique muscle). Finally, when injecting around the eyes, stay away from the superficial periorbital veins to avoid any bleeding. In addition, the lacrimal gland deeply behind the frontomalar apophysis (lateral supraorbital rim) should be avoided.

Botulinum neurotoxin type A in the mid and lower face and neck

Generally, due to the underlying anatomy, the mid and lower facial and neck areas are associated with volumetric changes versus the strictly dynamic changes more commonly seen in the upper face. Dynamic lines are not as prevalent as in the upper face; instead, there is more sagging, fat and skin atrophy, and laxity. Because of this, multimodal treatment regimens are often utilized. Autologous fat transfer is also an option to replace volume; however, it is a more complicated and time-consuming procedure (i.e. it is not a "lunch-time" procedure).

Superficial muscles present in the mid and lower face and neck include the nasalis, orbicularis oris, risorius, and platysma. Deeper muscles include the zygomaticus major and minor (under the orbicularis oculi, orbicularis oris, and the malar fat pad), levator labii superioris and levator labii superioris alaeque nasi (under the orbicularis oris, malar fat pad, and nasolabial folds), depressor anguli oris (oblique from the corner of the mouth (modiolus) down and laterally to the mandibular border, joining to the fibers of the platysma), and the mentalis (two parts, perpendicular from the mandibular symphysis, joining together and spreading to the skin, crossing all the fibrotic chin tissue like a lotus). Figure 3 depicts some of the lower face muscular structures (7).

General anatomical and treatment guidelines for the mid and lower face and neck are provided in Table II (1,2,4,7–10,12,14–16). When treating these areas, a number of anatomical features in addition to the musculature should be considered,

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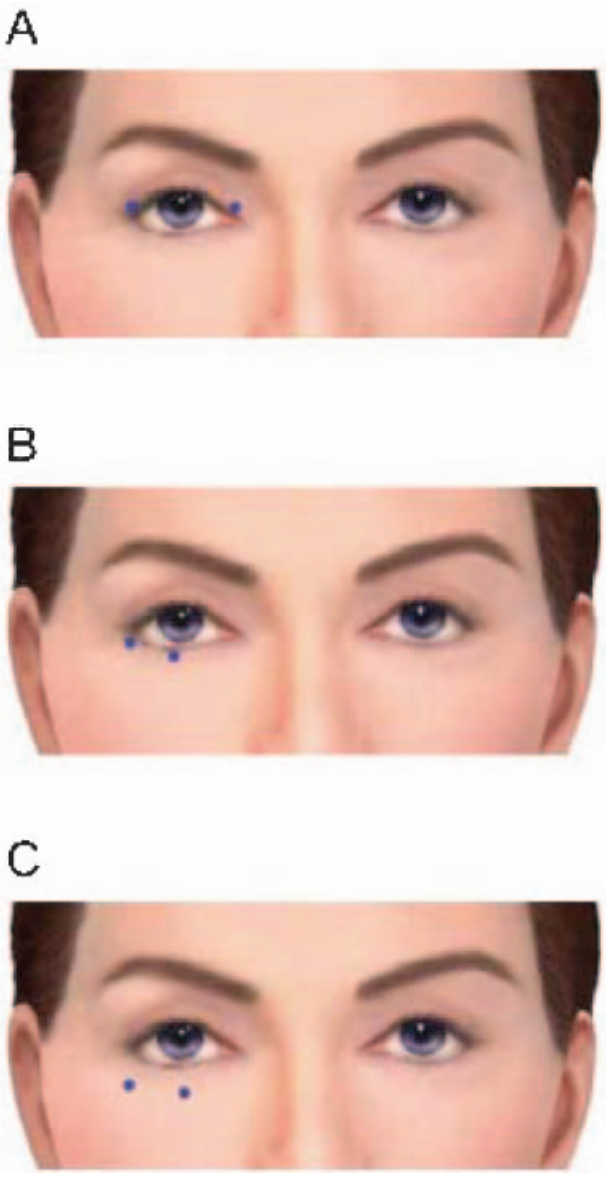


Figure 2. Suggested sites of injection of botulinum neurotoxin type A for (A) mild upper lid blepharoptosis; (B) mild to moderate lid fissure asymmetry; and (C) dynamic lower eyelid rhytids. Reprinted with permission from: Fagien S. Temporary management of upper lid ptosis, lid malposition, and eyelid fissure asymmetry with botulinum toxin type A. Plast Reconstr Surg. 2004;114:1892–902. Images courtesy of Steven Fagien, MD.

including the facial artery on the mandibular edge (at the anterior part of the masseter, a pulse can be felt with palpation); the malar fat pad, superficial to the zygomaticus muscle; and the thickness of the superficial subcutaneous fat at the jowls, mandibular line, and neck (note the yellow fat in Figure 3). In addition, the perioral skin covering the chin and lips is thicker than in other parts of the face; thus, injections must be deeper in this area, even though the targeted muscles are immediately subcutaneous (1,6,7).

Conclusion

In summary, botulinum neurotoxin type A is increasingly being utilized—with good results—beyond the traditional upper facial applications into additional upper, mid, and lower facial and neck applications. A comprehensive understanding of functional anatomy is crucial to maximizing outcomes in these more recently considered areas. In addition, while dosage and injection site guidelines and consensus recommendations, along with expert technical application, are invaluable, each patient needs to be thoroughly assessed and have a specific treatment plan developed, based on patient desires and a comprehensive aesthetic evaluation.

Discussion

The meeting on which this supplement is based allowed for a discussion period following each presentation. Here are highlights from the discussion that ensued after ''Facial Rejuvenation With Botulinum Neurotoxin: An Anatomical and Experiential Perspective''.

When injecting in the periorbital area, how far laterally do you inject?

Alastair Carruthers, MD: It is important to inject orbicularis oculi approximately 1.5 cm from the lateral commissure at the place where the muscle mass is maximal. If you stay too close to the commissure, you're in danger of creating a ridge lateral to the area of weakness that is quite unsightly.

Steven Fagien, MD: In most patients, I use the orbital rim as my landmark. I separate individual injection points from 0.5 to 1 cm. I inject small blebs in this region, and extend them cephalad and caudally, depending on the desired effect. I find that with this technique, I often don't have to extend out another ''row'' laterally; I also don't need to inject more medially. I'm depending, in part, on mechanical diffusion to homogeneously saturate the muscle (neuromuscular junction of the orbicularis oculi) in this region.

What dilution do you use to treat the periorbital area?

Steven Fagien, MD: I generally use a 4 cc dilution. You get used to delivering a result with your particular dilution preference and know how to deliver your dilution. Could I change to a 1 cc or 2 cc dilution? Yes. Is one dilution better than another? No, but in my hands, I know the nuances of my particular dilution.

Discuss the difference between treating vertical platysmal bands and horizontal necklace lines.

Steven Fagien, MD: Platysmal bands result from muscular activity (active lines), while

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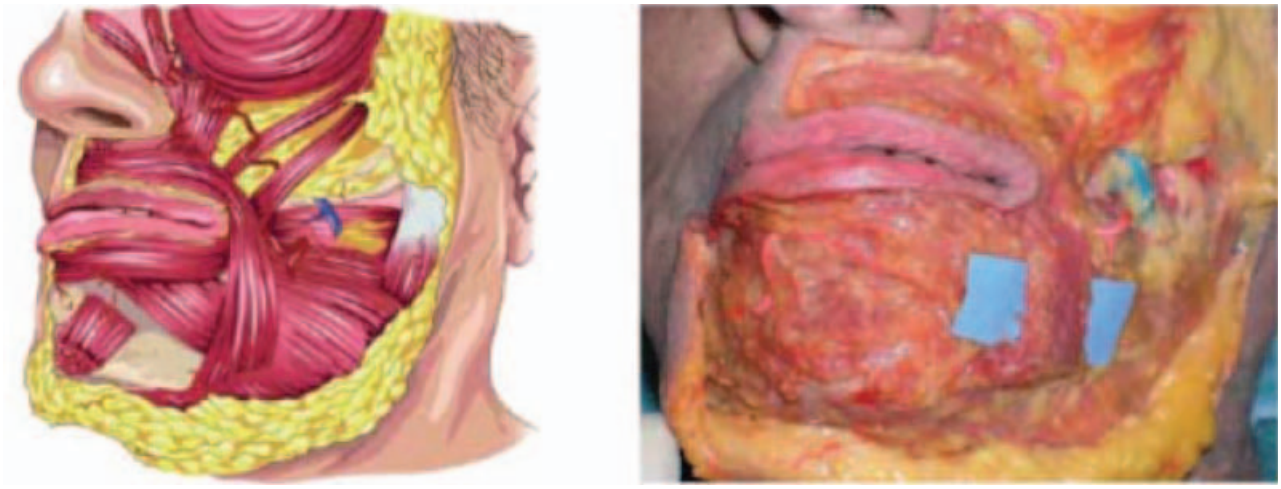


Figure 3. Muscles of the lower face shown in planar sketch and dissection view. Observe clockwise the midface and perioral mimic muscles (in red). The nose: *nasalis*; *levator labii superioris alaeque nasi*; *levator labii superioris*; *orbicularis oculi* (on top of the sketch); *zygomaticus major* (note the sinusoid of the facial artery, in dark red); *zygomaticus minor*; *risorius*, crossed by the facial vein (in blue); *depressor anguli oris* (DAO; above a blue rectangle of paper in the dissection view); *platysma* (lateral to the DAO); *depressor labii inferioralis* (medial to the DAO); *mentalis* (cut); *orbicularis oris* surrounding the pink lips. Note (in yellow) the subcutaneous fat and the thickness of the chin skin. Sketch by G. Blanchet; photograph, courtesy of L. Belhaouari: Belhaouari L, Gassia V. L'Art de la Toxine Botulique [in French]. Editions Arnette (Wolters Kluwer France): Rueil-Malmaison, France, 2006. Reproduced by kind permission of the Publisher (7).

Table II. Emerging botulinum neurotoxin type A applications: mid and lower face and neck (1,2,4,7–10,12,14–16).

Treatment area	Muscles involved	Anatomical considerations	Total BoNTA dose/injection sites	Injection technique	Notes
Bunny lines	Nasalis, procerus	Nasalis is a paramedian nasal muscle; fibers course in superomedial direction from origin on the maxilla over the nasal bridge to fuse with contralateral muscle and the aponeurosis of the procerus muscle; heavily vascularized area	2–5 U (nasalis) 1 U (procerus, if needed). 2–3 sites (1 per side for nasalis, 1 in midline for procerus)	Injections should be relatively superficial	Often treated in conjunction with the glabellar complex
Nasolabial folds	Levator labii superioris alaeque nasi,	Levator labii superioris alaeque nasi is the most medial of the mimetic muscles, found in the groove between the nasal bone and the maxilla.	3.75–15 U (most often 5–7.5 U)	Levator labii superioris alaeque nasi: 5 mm laterally and horizontally to the nostril (depth ~8 mm)	Generally recommended only for patients with ''gummy smiles''; even in this subgroup, many patients dissatisfied with alteration of smile mechanics; rejuvenation of this area may best be accomplished via other modalities (rhytidectomy, fillers)
	zygomaticus major	Zygomaticus elevates laterally the corner of the mouth and so it augments the nasolabial folds	1–3 U	Zygomaticus: upper insertion: 3 cm vertically down to the corner of the eyes (depth ~8 mm at the contact to the end of the malar bone); lower insertion: 1 cm laterally and superior to the mouth corner (depth ~6 mm)	
Vertical lip lines	Orbicularis oris	Orbicularis oris is a circular muscle with upper and lower lip contributions and deep and superficial fibers; upper lip portion inserts laterally on the maxilla opposite the lateral incisor tooth; lower lip portion originates on the mandible (lateral to the mentalis); both portions insert/fuse with the DAO and risorius at the oral commissures in the modiolus; contraction of the orbicularis oris deep fibers pulls the lips close to the teeth, whereas superficial fibers contract to project the lips together anteriorly	4–10 U 2–11 sites	Some practitioners inject the upper lip only (at the lip margin, or junction of red and white lip), others, the upper and lower lip; injections should be kept superficial and symmetrical; avoid midline	Often used in combination with laser resurfacing and/or fillers; desirable ''plumping'' of the lips can occur; overtreatment can result in dysfunction (speech impairments, trouble eating, etc.)

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Table II. Continued.

Treatment area	Muscles involved	Anatomical considerations	Total BoNTA dose/injection sites	Injection technique	Notes
Marionette lines	Depressor anguli oris (DAO)	DAO origin is oblique line of the mandible lateral and inferior to the depressor labii inferioris; inserts on the modiolus as it fuses with fibers from the risorius and orbicularis oris; fibers also intermingle with the platysma; contraction causes the corners of the mouth to turn down; marionette lines result from DAO hyperactivity at its medial insertion	4–6 U 2 sites	Each muscle should be injected 1 cm laterally and 2 cm vertically inferior to the angle of the mouth	Often used in combination with laser resurfacing, face-lifting, and/or fillers; overtreatment can result in dysfunction (speech impairments, trouble eating, etc.)
Pebbly chin	Mentalis	Mentalis originates inferiorly at the incisive fossa of the mandible in the mentum and travels vertically in the midline to fuse with the orbicularis oris; contraction causes lower lip eversion and skin wrinkles associated with an expression of doubt	2–6 U (women), 2–8 U (men) 1–2 sites	Inject low just below the prominence of the chin; injections should be angled upward and perpendicular to the chin	Often used in combination with fillers
Masseter reduction	Masseter	Upper insertion (2.5 cm long): zygoma apophysis (lateral part of the malar bone). Lower insertion (4.5 cm long): inferior mandibular border	50–60 U 10–12 injection sites	Very deep injection (1.5 cm) going through all facial tissues down to the bone; contact at the junction point (3 cm) vertically upward to the mandibular border and 3 cm forward or the posterior mandibular border	Sometimes used to reshape squared faces with hypertrophic masseteric muscles, NOT mandibular bony prominence, such as in some patients of Asian ethnicity (muscular hypertrophy must be the underlying cause for BoNTA to be efficacious)
Platysmal bands	Platysma	Platysma is a broad, thin muscle that spans the anterolateral aspect of the neck; originates from the superficial fascia overlying the upper chest, clavicle, and acromial regions, ascends in a superomedial direction in the neck to insert at 3 points (medially into the mentum; centrally into the periosteum of the ramus of the mandible; laterally to the orbicularis oris, DAO, risorius, and mentalis muscles and overlying subcutaneous tissue); contraction pulls skin overlying the clavicle upward and partially opens the mouth by pulling oral commissures downward and lateral	10–30 U (women), 10–40 U (men) 2–12 sites/band (women), 3–12 sites/band (men)	Grasp band with nondominant hand and inject into belly of muscle	Patient selection critical; should be used in patients who have retained skin elasticity and have minimal descent of submental fat; 2 bands/session usually treated

BoNTA=botulinum neurotoxin type A; DAO=depressor anguli oris.

necklace lines are more often static lines. Chemodenervation works much better on active lines, so results are usually much better with the improvement of platysmal bands versus reducing or eradicating necklace lines. As in most areas of the face, you need to discriminate between dynamic versus static.

What is your experience with using botulinum neurotoxin type A in the platysma?

Nicholas Lowe, MD: If the botulinum neurotoxin injections are placed properly along the platysma, the experience is good. The neck is an area where you have to be cautious with the total dose to avoid deeper laryngeal or

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pharyngeal side effects. The duration of effect can be longer in the platysma than in the upper face.

Steven Fagien, MD: Most people have had variable experiences. In the early studies, high doses were used—up to 300 U—and investigators observed a dramatic variability in results. Young patients can do much better with the appropriate treatments, because the integrity of their retaining ligaments is superb. By relaxing the platysmal contractions, the soft tissue of the neck more often lies closer to the desired neck contours. Another great subgroup, for similar reasons, are those who have had lower facial surgery and have residual banding. Botulinum neurotoxin type A, however, does not work well on every individual, and your evaluation will usually detect which individuals will benefit the most, as well as those who may be disappointed.

*Are there any instances where Dysport® (*Clostridium botulinum *type A toxin-haemagglutinin complex, Ipsen Limited, Slough, UK) is used instead of BOTOX in facial rejuvenation?*

Charles Finn, MD: Using two products with different characteristics and dilutions in the office may invite dosing errors. These products are not interchangeable, and the practitioner needs to exercise forethought in determining injection patterns and dosing for each anatomic area.

Nicholas Lowe, MD: I have used Dysport on the platysma because there may be increased diffusion, which can be of benefit in the platysma. I restrict the amount to a maximum of 100 U, and I have not observed laryngeal or pharyngeal side effects. This increased diffusion may be a disadvantage in the rest of the face, where we are trying to be precise and achieve specific results. I'll alternatively suggest Dysport in the glabella if a patient is not happy with the results achieved with BOTOX.

Hervé Raspaldo, MD: I would be concerned about diffusion in the neck. I would not recommend using Dysport in the neck. I will try Dysport in the forehead if a patient is not happy after multiple therapy sessions with BOTOX; I never use Dysport for a first session.

Sue Ellen Cox, MD: I'd be less likely to use Dysport in the lower face because of diffusion. It may be an option in the forehead and glabella if it is less expensive.

Joel Cohen, MD: The concept of using two toxins in the same patient concerns me. Some may favor Dysport for facilitating diffusion in the forehead or for axillary hyperhidrosis, while using BOTOX in the glabellar area or mid and lower face during the same treatment session. If a problem occurs when using two toxins in the same patient, you are not going to know which one is causing it. Then the patient would stay away from both, and you've lost an opportunity.

Nicholas Lowe, MD: For BOTOX, I currently use a dilution of 2.5 mL per 100 U vial, so that 0.1 mL equals 4 U. For Dysport, my dilution is 3.2 mL per 500 U vial, to yield 15.6 U per 0.1 mL, or about 4 U Dysport to 1 U BOTOX.

Disclosures

Dr Fagien is a consultant for Allergan, Inc.

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